NLP-Enhanced Compatibility Modeling for Social Matching: A Text-First Benchmark Using the OKCupid Dataset

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Abstract

This project update presents preliminary results for an NLP-enhanced compatibility modeling system targeting social matching platforms. We evaluate three text representation methods—TF-IDF vector space models, Word2Vec average pooling, and SBERT sentence embeddings (6)—on self-description essay text from the OKCupid dataset (59,946 profiles) (10). A complementary attribute-based compatibility benchmark is constructed from structured profile fields, and two classifiers are compared: a logistic regression baseline on simple demographic rules and a gradient boosting model on weighted soft-compatibility labels. Our key findings are: (1) SBERT produces the most discriminative similarity distributions for profile matching; (2) Word2Vec average pooling collapses toward a high-mean, low-variance distribution due to mean vector averaging (5), rendering it unsuitable for document-level retrieval without pooling refinement; and (3) orientation and age gap dominate attribute-based compatibility prediction, while religion and diet provide marginal signal due to high missingness rates. These results motivate a hybrid fusion architecture for the final system.

1 Introduction

Compatibility prediction in social matching platforms is conventionally driven by engagement signals—swipe rates, recency, and popularity—rather than measures of psychological or semantic alignment between users (1). This project investigates whether richer text-derived representations of user intent and personality can improve compatibility modeling, with the long-term goal of building a matching algorithm for a social app MVP.

The core hypothesis is that user-generated free-text profile descriptions encode latent compatibility signals that behavioral signals alone cannot capture. Specifically, we ask: do dense sentence embeddings derived from the pretrain-finetune paradigm (2) produce more useful similarity estimates than classical sparse representations such as TF-IDF (3), and how do these text signals interact with structured attribute compatibility features?

This update describes our data, the two preliminary experiments we have run, and the insights gained. Due to a data constraint discovered during preprocessing—essay text and structured profile attributes are decoupled across dataset rows, preventing direct fusion—we evaluate text-based and attribute-based signals independently and propose a fusion architecture for the final system.

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2 Related Work

2.1 Text Representation for Compatibility

The dominant paradigm for document similarity begins with the vector space model (VSM), in which each document is represented as a TF-IDF weighted bag-of-words vector and similarity is measured by cosine distance (3). While effective for topical retrieval, VSM suffers from the curse of dimensionality and lexical sparsity: two profiles discussing the same interests in different words receive near-zero similarity. Salton and Buckley (3) formalized IDF weighting as $\mathrm{idf}_t = \log(N/\mathrm{df}_t)$, which down-weights common terms and amplifies informative rare ones, but cannot bridge semantic gaps between synonymous expressions.

Neural word embeddings address sparsity by projecting words into dense low-dimensional spaces where semantic relationships are geometrically encoded. Mikolov et al. (4) introduced Word2Vec's skip-gram objective, which maximizes $\sum_{-c \le j \le c,\, j \ne 0} \log P(w_{t+j} \mid w_t)$ over a sliding context window. Document-level representations are commonly obtained by average pooling word vectors, but as we demonstrate, this approach collapses all profiles toward the centroid of the embedding space, producing uniformly high cosine similarities that are useless for ranking (5).

Sentence-level representation learning with SBERT (6) overcomes this by fine-tuning BERT (2) using a siamese network with a contrastive or cosine-similarity loss over labeled sentence pairs. The resulting embeddings are optimized for semantic similarity tasks and produce well-distributed pairwise scores suitable for retrieval and ranking.

2.2 Attribute-Based and Hybrid Matching

Traditional matching algorithms use attribute filtering (hard dealbreakers on religion, children preferences, orientation) combined with engagement-based ranking (1). Collaborative filtering approaches model user–item interactions via matrix factorization (7), but suffer from cold-start problems for new users. LightFM (7) partially addresses cold-start by incorporating user and item side features into the factorization objective; in our setting, NLP embeddings serve as these side features.

Emotion and affect modeling provides a complementary compatibility signal beyond semantic content. GoEmotions (8) provides fine-grained emotion labels that can be used to derive per-profile affect vectors. In our preliminary system we substitute video-based profile content with text-derived sentiment signals using a RoBERTa-based classifier (9), as the available dataset contains no video.

3 Proposed Framework

3.1 Dataset and Preprocessing

We use the OKCupid Profiles dataset (10) ($N = 59{,}946$ profiles) containing structured demographic fields (age, sex, orientation, religion, diet, education) and ten free-text essay responses. A critical preprocessing note: per the dataset documentation, essay rows have been randomized independently of structured profile rows to protect user privacy. Consequently, text and structured features cannot be joined at the user level, and we evaluate each signal stream independently.

For text experiments we concatenate `essay0` (self-summary; 90.8% filled, mean 635 characters) and `essay4` (favorites; 82.4% filled, mean 584 characters). For attribute experiments we use the structured fields directly, excluding income (75th percentile reported as -1, indicating widespread non-disclosure).

3.2 Text Representation Pipeline

We evaluate three representations spanning the sparse-to-dense progression covered in course Lectures 3–10.

TF-IDF. Each profile text is represented as a $|V|$-dimensional vector with TF-IDF weights (vocabulary capped at 10,000 terms, sublinear TF scaling, English stopword removal). Pairwise cosine similarity is $\cos(\mathbf{q}, \mathbf{d}) = (\mathbf{q} \cdot \mathbf{d})/(\|\mathbf{q}\|\|\mathbf{d}\|)$.

Word2Vec average pooling. We train a skip-gram model (dim = 100, window = 5, min count = 2) on the profile corpus and represent each document as the mean of its word vectors.

SBERT. We encode profile texts with `all-MiniLM-L6-v2`, a distilled BERT variant fine-tuned on over one billion sentence pairs (6), producing 384-dimensional embeddings optimized for cosine similarity. GPU acceleration (NVIDIA GTX 1650 Ti, CUDA 11.8) reduces encoding time for 2,000 profiles to under 30 seconds.

3.3 Attribute Compatibility Benchmark

We construct a pseudo-labeled benchmark of 10,000 sampled profile pairs under two label schemes.

Baseline label. A pair is labeled compatible (1) if both profiles share the same sexual orientation *and* have an age gap of at most 10 years.

Soft compatibility score. A weighted sum across five attributes—orientation match (weight 3.0), age-gap decay (weight 2.0), religion match (weight 1.5), diet match (weight 1.0), and education proximity (weight 1.0)—normalized to $[0, 1]$. Pairs scoring above 0.55 are labeled compatible. Religion values are normalized by stripping commitment qualifiers (e.g., "agnosticism but not too serious about it" $\rightarrow$ "agnosticism").

3.4 Planned Full System

The final system will integrate these signals into a hybrid architecture: (1) SBERT embeddings and RoBERTa-derived sentiment vectors as NLP side features; (2) structured attribute compatibility scores as preference filters; and (3) a LightFM (7) collaborative filtering layer trained on simulated or real interaction data, with NLP features addressing cold-start. An XGBoost re-ranker (12) will combine all signals for final candidate scoring.

4 Preliminary Experiment Results

4.1 Experiment A: Semantic Similarity Comparison

We sample 2,000 profiles and compute all ≈2 million pairwise cosine similarities for each method. Table 1 summarizes the results; Figure 1 shows the full distributions.

Table 1: Pairwise cosine similarity statistics ($n = 2{,}000$ profiles, `essay0` + `essay4`).

Method	Dim	Mean	Std	Max
TF-IDF	10,000	0.033	0.024	0.385
Word2Vec (avg pool)	100	0.954	0.047	0.999
SBERT (MiniLM-L6-v2)	384	0.352	0.140	0.823

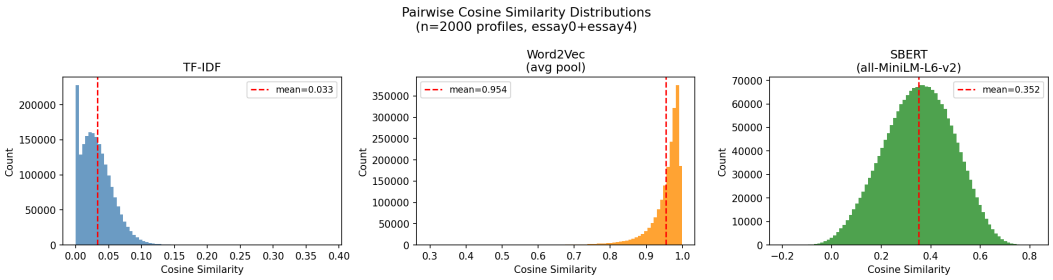

Figure 1: Pairwise cosine similarity distributions for TF-IDF (left), Word2Vec avg pooling (center), and SBERT (right). $n = 2{,}000$ profiles, `essay0` + `essay4`.

The distributions reveal qualitatively different behaviors. TF-IDF produces a right-skewed, near-zero distribution (mean = 0.033), consistent with high-dimensional sparsity: most profile pairs share almost no exact vocabulary, reflecting the limitations of bag-of-words representations (3).

Word2Vec average pooling yields a pathologically collapsed distribution (mean = 0.954, std = 0.047), with virtually all pairs scoring above 0.85. When hundreds of word vectors are averaged, the resulting document vectors converge toward the centroid of the embedding space, erasing document-specific signal (5). This directly illustrates why average pooling is insufficient for document-level retrieval without pooling refinement such as SIF weighting (5).

SBERT produces the most useful distribution: approximately normal, centered near 0.35, with standard deviation 0.14 spanning $[-0.2,\ 0.8]$. This spread means SBERT can meaningfully rank profiles from least to most semantically similar—a prerequisite for any useful retrieval or recommendation system.

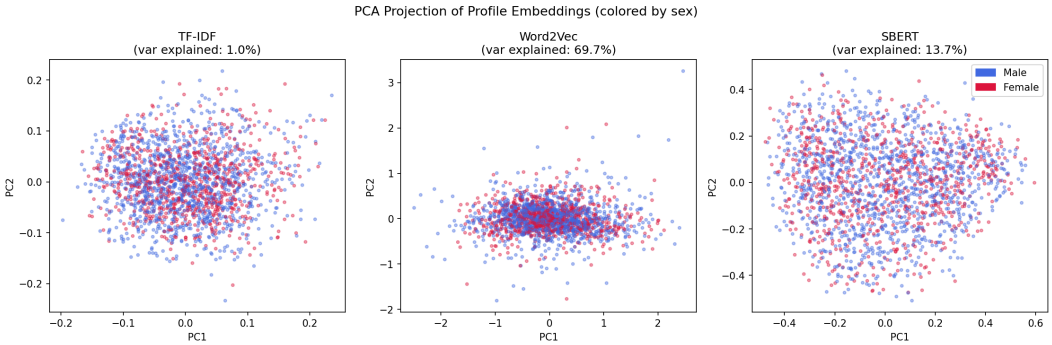

Figure 2: PCA projections of profile embeddings colored by reported sex. Explained variance: TF-IDF 1.0%, Word2Vec 69.7%, SBERT 13.7%.

The PCA projections (Figure 2) corroborate these findings. TF-IDF's 1.0% explained variance reflects extreme high-dimensionality. Word2Vec's 69.7% confirms that the 100-dimensional space has collapsed into a low-variance subspace. SBERT's 13.7% is appropriate for a 384-dimensional space with genuine semantic variation—information is distributed across many dimensions rather than concentrated in a few, which is desirable for retrieval. Notably, no method cleanly separates profiles by sex in 2D, suggesting writing style is not strongly gender-determined at the population level.

4.2 Experiment B: Attribute Compatibility Classification

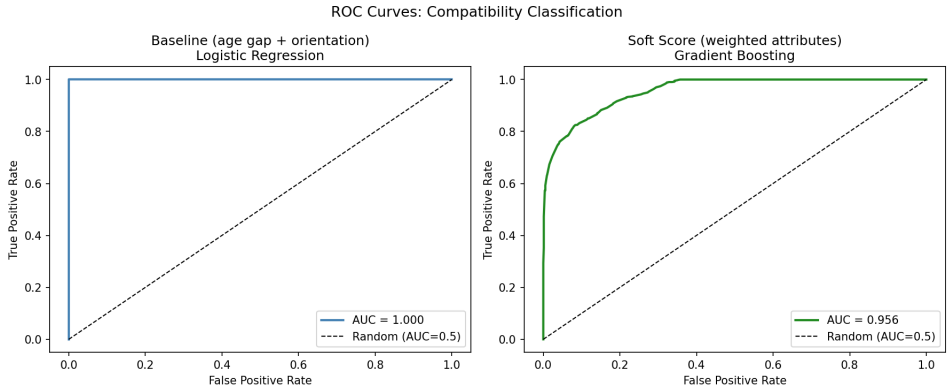

Figure 3: ROC curves for compatibility classification. *Left:* logistic regression on baseline labels (AUC = 1.000). *Right:* gradient boosting on soft compatibility labels (AUC = 0.956).

The baseline logistic regression achieves AUC = 1.000, which is an artifact of label construction: the model is trained on the same two features (orientation match, age gap) used to define the baseline label, making the classification task trivially learnable. This motivates the soft-label approach.

The gradient boosting model on soft labels achieves AUC = 0.956. The model successfully learns a non-trivial combination of four features to predict multi-attribute weighted compatibility. This result

should nonetheless be interpreted cautiously: because the features are the same attributes used to define the labels, the task is structured prediction of our own scoring function rather than prediction of true human compatibility.

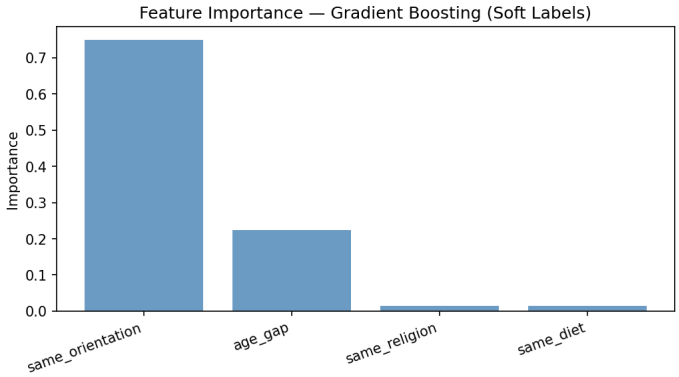

Figure 4: Gradient boosting feature importances on soft compatibility labels.

Feature importance analysis (Figure 4) reveals that orientation (importance 0.75) and age gap (0.23) dominate, while religion and diet contribute near-zero signal. This result is driven primarily by data missingness: religion is absent for 33.7% of profiles and diet for 40.7%, meaning these features are simply unavailable for a large fraction of pairs. This highlights the challenge of building compatibility benchmarks from datasets with high structured-field missingness.

4.3 Key Insights

Three actionable insights emerge. First, SBERT is the clear choice for text-based profile similarity: its well-distributed embeddings provide the discriminative spread needed for ranking, and GPU-accelerated encoding is practical on consumer hardware. Second, Word2Vec average pooling must not be used for document retrieval without pooling refinement; SIF-weighted pooling (5) or a pretrained sentence encoder are natural replacements. Third, attribute-based compatibility features have high missingness that must be addressed through imputation or Jaccard-style partial matching in the final system.

5 Next Steps

The following experiments are planned for the final system: (1) RoBERTa-based (9) sentiment encoding of essay text to construct per-profile affect vectors; (2) evaluation of SBERT similarity scores against the soft compatibility labels as a cross-signal correlation study; (3) integration of NLP embeddings as side features into LightFM (7) trained on simulated interaction data; and (4) an XGBoost re-ranker (12) combining semantic similarity, attribute compatibility, and activity signals from the Dating App Behavior dataset (11).

References

[1] Tyson, G., Perta, V. C., Haddadi, H., & Seto, M. C. (2016). A first look at user activity on Tinder. In *Proceedings of the 2016 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, pp. 461–466. `https://doi.org/10.1109/ASONAM.2016.7752275`

[2] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*. `https://arxiv.org/abs/1810.04805`

[3] Salton, G., & Buckley, C. (1988). Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5), 513–523.

[4] Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *arXiv preprint arXiv:1310.4546*. `https://arxiv.org/abs/1310.4546`

[5] Arora, S., Liang, Y., & Ma, T. (2017). A simple but tough-to-beat baseline for sentence embeddings. In *International Conference on Learning Representations (ICLR)*. https://api.semanticscholar.org/CorpusID:64908139

[6] Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *arXiv preprint arXiv:1908.10084*. https://arxiv.org/abs/1908.10084

[7] Kula, M. (2015). Metadata embeddings for user and item cold-start recommendations. *arXiv preprint arXiv:1507.08439*. https://arxiv.org/abs/1507.08439

[8] Demszky, D., Movshovitz-Attias, D., Ko, J., Cowen, A., Nemade, G., & Ravi, S. (2020). GoEmotions: A dataset of fine-grained emotions. *arXiv preprint arXiv:2005.00547*. https://arxiv.org/abs/2005.00547

[9] Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*. https://arxiv.org/abs/1907.11692

[10] Kirkegaard, E. O. W., & Bjerrekaer, J. D. (2016). The OKCupid dataset: A very large public dataset of dating site users. *Open Differential Psychology*. https://doi.org/10.26775/ODP.2016.11.03

[11] Nisar, K. (2024). Dating App Behavior Dataset. *Kaggle*. https://www.kaggle.com/datasets/keyushnisar/dating-app-behavior-dataset

[12] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, pp. 785–794. https://doi.org/10.1145/2939672.2939785